

Lecture 10: Thermal Properties of Matter – Supplementary Notes

Hope Donglo, Office 24, Department of Physics, University of Ghana

These notes support Lecture 10. They contain background that was moved out of the contact time. Sections marked (extension) are not assessed.

1 Pressure units and gauge pressure

1 atm = 101.325 kPa = 760 mmHg $\approx$ 14.7 psi and 1 bar = 10^5 Pa. Tyre gauges, cylinder gauges and blood-pressure monitors read the *gauge* pressure, $p - p_{\text{atm}}$. The ideal-gas equation and the kinetic theory refer to the absolute pressure, the force per area exerted by the gas itself.

2 The ideal gas as a model, and its limits (extension)

The ideal gas has point molecules and no forces between them. Real molecules have a finite size and attract one another at a distance. The van der Waals equation corrects for both:

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT,$$

where b is the volume excluded by the molecules (per mole) and a measures the attraction. For nitrogen, $a \approx 0.137 \text{ Pa m}^6/\text{mol}^2$ and $b \approx 3.9 \times 10^{-5} \text{ m}^3/\text{mol}$. For 1 mol in 1.0 L at 300 K the ideal-gas pressure is $2.49 \times 10^6 \text{ Pa}$ and the van der Waals pressure is $2.46 \times 10^6 \text{ Pa}$, about 1.5% lower; in 10 L the difference is 0.16%. The ideal-gas equation is therefore a very good model at low density and a poor one near condensation, where the isotherms of Contact 2 develop a flat liquid-vapour section.

3 The pressure derivation in more detail

The cube is a convenience: the result $p = \frac{1}{3}(N/V)m(v^2)_{\text{av}}$ holds for a container of any shape. Collisions between molecules do not change the result: they exchange momentum and energy between molecules, but leave the average of v_x^2 unchanged because the number of molecules at each speed stays the same in equilibrium. Using only the time $2L/v_x$ between wall collisions neglects those collisions, and the correct result is the same. The factor $\frac{1}{3}$ appears because the motion is random: $(v_x^2)_{\text{av}} = (v_y^2)_{\text{av}} = (v_z^2)_{\text{av}}$ and their sum is $(v^2)_{\text{av}}$.

4 The Maxwell speed distribution (extension)

The fraction of molecules with speeds between v and $v + dv$ is $f(v) dv$ with

$$f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT}.$$

The three characteristic speeds are

$$v_{\text{mp}} = \sqrt{\frac{2kT}{m}}, \quad v_{\text{av}} = \sqrt{\frac{8kT}{\pi m}}, \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}},$$

in the ratio 1 : 1.128 : 1.225. For N_2 at 300 K: $v_{\text{mp}} = 422$, $v_{\text{av}} = 476$ and $v_{\text{rms}} = 517$ m/s. The factor $e^{-mv^2/2kT}$ falls off rapidly at high speed but never reaches zero, which is the tail referred to in the Earth-atmosphere extension.

5 Mean free path (extension)

A molecule of radius r travels, on average, a distance $\lambda = \frac{kT}{4\pi\sqrt{2}r^2p}$ between collisions. For nitrogen at 20°C and 1 atm, with $r \approx 2.0 \times 10^{-10}$ m, $\lambda \approx 5.6 \times 10^{-8}$ m, about 150 molecular diameters; the time between collisions is of order 10^{-10} s. At low pressure λ is large, which is why a vacuum chamber behaves almost as the kinetic model assumes.

6 Why latent heat exists

In a solid the molecules are held in a regular arrangement by attractive forces; in a liquid they are still close but free to move past each other; in a gas they are far apart. Melting needs energy to loosen the arrangement, and vaporization needs much more to separate the molecules completely, so $L_v > L_f$. During the change the *average kinetic energy per molecule*, and hence the temperature, is unchanged ($K = \frac{3}{2}kT$); the energy goes into the potential energy of the molecular arrangement. This distinction between kinetic and potential energy of the molecules leads to the internal energy of Lecture 11.

7 Critical point, triple point and the water phase diagram

For water the triple point is 273.16 K and 610 Pa (0.006 atm), and the critical point is 647.4 K and 2.21×10^7 Pa (218 atm). For carbon dioxide the triple point is 216.55 K and 5.17×10^5 Pa (5.1 atm), above atmospheric pressure, so solid CO_2 sublimates at 1 atm. Water is unusual: ice is less dense than liquid water, so the fusion curve slopes slightly backward (higher pressure lowers the melting temperature). Replotting the isotherms of a real gas as a surface over the p, V, T axes gives the pVT surface, whose projections are the pV and pT diagrams.

8 Equipartition (preview of Lecture 11)

The result $K = \frac{3}{2}kT$ is the energy of three independent translational motions: $\frac{1}{2}kT$ for each. The equipartition theorem states that each independent quadratic term in the energy of a molecule has an average of $\frac{1}{2}kT$ in equilibrium. A monatomic gas has only the three translational terms, so its energy per mole is $\frac{3}{2}RT$; a diatomic molecule at room temperature also has two rotational terms, giving $\frac{5}{2}RT$. These numbers set the molar heat capacities of gases, which we meet with the first law.